Gulf merchant platform production roadmap

GCC multi sector strategy and gated implementation plan

Research 27 October 2026

Decision

Target the six GCC markets through shared loyalty and service-progress capabilities, with distinct retail, food-service and express-service modules. Select the first paid country-sector-connector cell before implementation commitment. Treat messaging, issuer ownership and source coverage as release dependencies.

Production package

This report supplies the market decision, scope, architecture, contracts, economics, validation gates and production requirements. Companion files contain the dependency-linked backlog, OpenAPI proposal, editable financial model and evidence register.

All implementation targets and financial drivers are proposals or assumptions. Source-backed facts are identified in the evidence register. No live app or merchant outcome is represented as completed.

Contents

1 Executive decision 3
2 Evidence and decision rules 3
3 Gulf market sectors and competition 4
4 Shared platform and initial module scope 10
5 Order handoff customer and staff journey 10
6 Integration choices and the first technical gate 11
7 Notifications and Wallet constraints 12
8 Brand and experience system 13
9 Architecture and technical decisions 14
10 Security privacy and reliability baseline 15
11 Measurement and experiments 16
12 Business model and financial interpretation 17
13 Roadmap and release gates 19
14 Distribution and operating model 20
15 Innovation portfolio with decision triggers 21
16 Acceptance and decisions before production 22
17 Technical specification appendix 23
18 Evidence register 31

1 Executive decision

Build a web-first merchant platform for the six GCC markets with shared Wallet loyalty, consent, notifications, integrations and analytics, plus optional order-pickup and service-queue modules. Serve retail, food service, express services and suitable appointment-led businesses through distinct workflow packs. Select the first country-sector-connector cell through paid evidence. Keep existing merchant systems authoritative for sales, calendars and entitlements; give customers useful status without forced registration.

The proposed product has two different jobs. Pickup coordination addresses the current visit: order visibility, reliable readiness, orderly collection and operational measurement. Loyalty addresses the next visit: a clear reward, verifiable earning and affordable fulfillment. Both can share purchase events, but each must prove its value separately. Gamification is an experiment inside loyalty, not a reason to delay reliable pickup operation.

Commercial judgment: 7 out of 10 for a focused merchant workflow with paid validation; substantially weaker if the entire regional portfolio is built before proving one country-sector-connector cell. Incumbents already supply KDS, customer displays, messaging and loyalty. The opportunity is a simpler missing workflow for a specific group of merchants, not a new category of QR technology.

Proceed through evidence gates. The first investment is merchant access, a receipt and printer capability test, and an instrumented paid pilot. Do not commit the full production budget until order coverage, staff adoption, notification economics and willingness to pay are demonstrated. If integration is blocked or the cafe's existing POS solves the problem adequately, stop that connector or that customer segment.

The target is GCC-wide and multi-sector. Founder access, legal entity, available team, merchant contracts and budget remain unestablished. UAE order pickup is the detailed worked case already researched; it is not the sole market, the only sector or a predetermined launch choice. Country and sector selection now precede implementation commitment. Gulf opportunities and country dependencies are assessed in the regional section.

2 Evidence and decision rules

Four labels govern this baseline. A fact is supported by a cited government, platform, research or first-party vendor source. A vendor claim establishes what a vendor advertises, not the outcome its customers achieve. An assumption is an editable planning input. A proposal or experiment is a decision awaiting validation. Forecasts, effort ranges, SLOs and success thresholds below are proposals, not measured performance.

Sources carry stable identifiers in the evidence register. Current commercial terms need reconfirmation before procurement. Source conflicts are retained where they change a decision. In particular, Meta's October 2026 developer pricing supersedes older free-message wording; restaurant licence counts are not cafe counts; native receipt modification remains unverified; and Apple certificate ownership is a contract gate.

No merchant interviews, paid sales, POS sandbox implementation, printer tests, physical-device Wallet tests, production security review or live load tests have been performed for this roadmap. Existing research supports feasibility and constraints. It does not establish product-market fit or a positive merchant ROI.

Evidence gates require saved outputs: interview notes with consent, redacted transaction exports, printer specimens, sandbox event traces, signed commercial quotes, actual pilot payments and repeatable acceptance results. A green checklist without its evidence is not a release decision.

3 Gulf market sectors and competition

Gulf market scope

The target market is the six GCC states: Bahrain, Kuwait, Oman, Qatar, Saudi Arabia and the United Arab Emirates. The product serves several kinds of merchant, not only cafes. GCC membership defines the geographical study; it does not create one privacy, tax, messaging, payments or merchant-onboarding jurisdiction. [G01]

The proposed shared platform contains merchant groups, legal entities, brands, branches, staff identity, customer consent, integrations, verified purchase facts, Wallet loyalty, notification routing, operational events and reporting. Order pickup and service queues are separate optional modules. Appointment calendars, membership entitlements, menus, inventory, acquiring and medical records remain external systems of record unless a later business case authorizes expansion.

The first country-sector-connector combination is selected by Gate 0 evidence. UAE cafe/QSR pickup is a researched reference case, not a restriction on the business. Saudi specialty-retail collection is a strong second discovery candidate. Run inexpensive interviews in both if access permits, but commit integration engineering to the cell that supplies paid demand, workable data and the lowest reliable activation cost. Neither has yet supplied those proofs.

Sector selection

Segment	Initial fit judgment	Useful module and boundary
Cafe and quick-service pickup	8 out of 10 candidate	Order status, ready/collected workflow, optional alerts and rewards. Integrate POS/KDS; do not rebuild kitchen routing or menu ordering.
Specialty retail and click and collect	8 out of 10 candidate	Ready, arrival and collection proof linked to an existing commerce order. Inventory availability, stock reservation and refunds stay authoritative upstream.
Express car wash	7 out of 10 candidate	Waiting, called, in-service and done, plus repeat-visit rewards. A wash queue is not a workshop-management system; vehicle plates are personal-data considerations.
Barbers and salons	6 out of 10 for loyalty alone; 5 for a new scheduling suite	Wallet loyalty and arrival/queue adapter can fit. Preserve the existing calendar, staff availability, deposits and rescheduling rules.
Boutique fitness	5 out of 10 pending integration	Verified attendance and optional recognition. A paid membership, class capacity, access right and loyalty stamp are different facts.
Clinics and health services	3 out of 10 default	Generic queuing can look similar, but patient identity, urgency and health data require a separate safety/privacy programme. Disabled in the general merchant release.

Scores are proposed product judgments, not measured market indices. They consider transaction cadence, staff pain, supplier overlap, integration depth and consequence of a wrong status. Test value per sector rather than transferring coffee margins or waiting times to every business.

Regional market evidence

Saudi GASTAT reports 1.022 million active enterprises in 2024, including broad wholesale/retail and accommodation/food categories. Those figures establish business breadth, not addressable stores or adoption of pickup tracking. Its retail operating-revenue statistics likewise cannot establish software willingness to pay. Use current activity-code and branch-level records when constructing a Saudi serviceable market. [G02] [G03]

Saudi commerce competition includes Salla and Zid, whose first-party offers already combine substantial order, storefront and physical-retail capabilities. Fresha supplies appointment/client workflows for beauty and wellness; its global page does not establish a local GCC price. Qminder and Queberry compete on queue operations, messaging and analytics. Integration rights and incremental value must be checked against what the actual merchant already enables. [G08] [G09] [G10] [G11] [G12]

Qatar MCIT's SMEs Go Digital programme reports actual SME onboarding, provider matchmaking and digital-solution adoption. This is a possible partner route, not product-demand evidence. Bahrain's official digital-economy strategy includes SME digital solutions. Oman's e-commerce strategy includes SME enablement, payments, logistics and skills. These are useful distribution contexts; none supplies a defensible target-market count for this product. [G05] [G06] [G07]

Market	Discovery and distribution hypothesis	Evidence required before local release
Saudi Arabia	Specialty retail collection through commerce/POS partners; QSR and express services also eligible	Named merchant buyer, supported connector, Arabic operation, data-transfer and tax-QR decisions, permitted channel and actual paid price
UAE	Owner-led pickup or repeat-service pilots; documented Foodics discovery path	One local operating cell, receipt/guest-order coverage, sender/issuer ownership and measured support burden
Qatar	SME programme and integrator-led retail or service workflow	Partner eligibility, merchant access, actual technology stack and local contracting economics
Oman	Retail or express-service pilot through a local digital-commerce integrator	Local customer workflow, messaging availability, OMR billing precision and data-processing requirements
Bahrain	Compact partner-led discovery for retail, QSR or repeat services	Target activity/location denominator, enabled incumbent functions, local price and support arrangement
Kuwait	Buyer and channel discovery before committing a connector	Merchant interviews, local POS/commerce mix, sender and privacy scope, contracting and collection requirements

The sequence is an option portfolio rather than six simultaneous sales launches. A country becomes active only with a responsible owner and its evidence pack. Lack of a measured Kuwait segment denominator is uncertainty, not evidence against that market. Do not fill gaps with another GCC state's adoption rate.

Country production packs

Country	Verified regulatory anchor	Production implication requiring local assessment
Saudi Arabia	SDAIA Personal Data Protection Law and transfer requirements; ZATCA e-invoicing requirements; CST SMS framework	Map controller/processor roles and actual transfers. Tracking QR must remain distinct from the merchant's required fiscal invoice QR. Confirm sender/provider permissions and chosen POS responsibilities. [C01] [C02] [C03] [C04]
UAE	Federal PDPL with separate DIFC and ADGM regimes; TDRA messaging controls	Select regime by actual entity and flow, document international access, and validate sender/issuer and commercial account eligibility. [P11] [P19] [P20] [P21]
Oman	Personal Data Protection Law under Royal Decree 6/2022 and Executive Regulation 34/2024	Assess processing/permit triggers, marketing, overseas processors and the data classes actually used. [C05]
Bahrain	Personal Data Protection Law framework and transfer-adequacy/authorization orders	Verify destination adequacy or required safeguards/approval for the real hosting and support arrangement. [C06]
Qatar	Law 13/2016 and NCSA regulated-entity guidance	Map electronic-data processing, notices, rights, marketing and processor obligations; determine country-specific incident and transfer handling. [C07]
Kuwait	CITRA Data Privacy Protection Regulation under Decision 26/2024	Assess its scope and other applicable sector/consumer rules; do not call it a blanket GCC-style omnibus PDPL. [C08]

These anchors are a requirements map, not legal clearance. Fiscal QR generation and tax invoices remain with the merchant's compliant system. Never replace or cover a required Saudi invoice QR with our tracking QR. Keeping the two links separate is a product acceptance condition, not a promise that our software certifies the whole invoice. [C03]

Separate two payment flows. Our company collecting a SaaS subscription from a foreign merchant is not the same as settling that merchant's customer sales. V1 reads payment facts from the merchant's existing POS and does not custody customer money. A future checkout or marketplace-settlement feature needs local PSP/acquirer and regulatory review. Stripe currently lists UAE for direct account support among GCC countries; this does not itself prohibit lawful cross-border SaaS sales, and it does not authorize us to aggregate unrelated GCC merchants' sales through one account. Actual entity, tax, account and payout terms govern each arrangement. [C09] [C10] [P16]

Localization and channel economics

Market	Local currency and decimal minor units	Meta utility list rate in AED
Saudi Arabia	SAR with 2 decimals	0.0393
UAE	AED with 2 decimals	0.0576
Qatar	QAR with 2 decimals	0.0440
Oman	OMR with 3 decimals	0.0907
Bahrain	BHD with 3 decimals	0.0334 using Rest of Middle East mapping
Kuwait	KWD with 3 decimals	0.1615

Currency exponents follow the ISO 4217 maintenance source. Message rates are the archived Meta AED list rates effective 1 October 2026, before tax/provider charges and relevant tiers. They follow recipient destination markets, not simply the cafe or shop's country. Utility and service have equal values in this edition but remain separate categories. An AED0.10 landed-cost example is not a viable cost assumption for a Kuwait utility message whose Meta base alone is higher. [C14] [C15] [A04]

Store purchase amounts as integer minor units with currency; use precise decimal provider rates and aggregate before invoice rounding. Do not round an AED0.0576 unit rate to two decimals per message or assume every currency uses fils with exponent two. Tenant ledger currency and valid currency conversions are server-controlled. A UAE worked case does not supply SAR, KWD or OMR pricing.

Arabic/English and RTL are shared requirements. Branch timezone is configured, not one hard-coded Gulf offset: Saudi, Qatar, Bahrain and Kuwait currently use UTC+3; UAE and Oman UTC+4. Store UTC event time with an IANA branch timezone and test locale-specific date, number and phone handling. Names, order codes and phone numbers need bidi isolation and merchant-approved copy.

Apple/Google Wallet pass distribution is distinct from Apple Pay/Google Pay payment acceptance. Candidate country availability does not establish issuer approval, every pass feature, local bank support or device compatibility. Run local account/device tests per launch cell. Health data and sensitive/private passes require separate provider and regulatory reviews; no medical-compliance claim is included. [C11] [C12] [C13] [C16]

Shared architecture and module contracts

Add merchant group, legal entity, brand and branch as distinct concepts. The group can span markets; a branch is linked to its operating entity, country, timezone, currency, enabled modules, sender assets and applicable data policy. Staff access stays tenant and location scoped. Billing seller identity, merchant identity and Wallet issuer identity are explicit fields, not inferred from a logo.

The shared loyalty engine should consume a verified purchase fact, not assume every earn comes from a cafe order. A purchase fact carries tenant, location, source transaction ID, currency, eligible value, paid/refund state and policy version; an order or service adapter establishes its provenance. Keep one-claim uniqueness at the purchase/program boundary. An attendance event alone is not a paid membership or a qualifying sale.

Use separate bounded workflows: order handoff progresses received, preparing, ready and collected; service queues progress waiting, called, in service and completed, with left/cancelled exceptions; appointment arrival links to the external booking. Share event envelopes and authorization, not one

arbitrary configurable state machine for every industry. Tenant settings may select supported workflow templates and thresholds; they do not inject arbitrary executable rules.

The core API proposal and the 75-item reference backlog implement the initial order-handoff path. A separate 24-item GCC expansion backlog covers country/entity configuration, currency precision, the purchase-fact boundary, service queues, retail adapters, car wash, appointment arrival, fitness entitlements and country release packs. These are 99 candidate delivery items in total, not 99 mandatory launch requirements.

The additional portfolio is estimated at 492 to 800 engineering hours plus 176 to 308 non-engineering hours if selected. Do not add that entire amount automatically to the UAE order-case budget: choose the required core-enabling items early and activate sector/country items only after a paid transfer case. X001 to X012 contain selection and shared-country foundations; relevant hierarchy, currency and purchase-contract decisions precede schema freeze, even when their full implementation follows the first pilot.

The financial workbook remains an explicitly labelled UAE order-handoff example with an optional loyalty layer. It is not a GCC-wide forecast or the complete cost of all country/sector modules. Reprice each active cell using local merchant margins, order/service cadence, support, destination message mix, issuer arrangements and applicable taxes. Candidate sectors can choose loyalty alone, pickup alone, service queue alone or a justified combination.

UAE order handoff reference case

Dubai DET reported almost 1,200 new restaurant licences in 2024. Dubai Municipality reported approximately 2,336 new food establishments in the first half of 2025. These figures describe different categories and periods; neither is a count of addressable cafes. Use them as evidence of sector activity, not as the denominator for a market-share forecast. [M01] [M02]

In this reference case, the buyer is an owner or operations manager at a one-to-three-site cafe or takeaway operator. The operational filter is counter pickup, measurable busy periods, recurring status questions or collection confusion, and a decision-maker able to authorize a short pilot. Preferred initial customer has a cloud POS with a feasible event or receipt path. Exclude table-service operations, delivery aggregation, very low-volume venues and enterprise procurement initially.

The problem must be visible during a rush. Ask how often customers interrupt staff, which orders remain uncollected, whether drinks are remade, what existing displays or pagers accomplish, and which metrics the owner already trusts. A cafe using an adequate built-in notifier has little reason to add another subscription.

Competitive options

Alternative	Verified position or public price	Consequence for our offer
Foodics	UAE page advertises AED199 Starter, AED375 Basic and AED556 Advanced per month, billed annually. Exact feature entitlements need a quote.	Closest integration candidate and an incumbent substitute. Additional API licensing may change total merchant cost. [M06] [M07]
Syrve	UAE page lists AED250, AED350 and AED450 per till monthly before VAT; annual options differ.	Customer displays, messaging and loyalty can already exist inside a wider POS suite. Confirm enabled modules before pitching. [M09]
Oracle Simphony	Transaction, BI and extensibility interfaces; partner process documented. No comparable public UAE quote verified.	Consider after demand from a chain justifies partner and procurement work. [M08]
Grubtech and Deliverect	Order operations and integration platforms; reviewed commercial offers require quotes.	Avoid duplicating delivery aggregation and full kitchen orchestration. [M10] [M11]
Queberry	Queue management, digital signage and WhatsApp queueing advertised.	Competes for the same customer-flow budget even with a different workflow. [M12]
Stampli UAE	AED69, AED179 and AED369 monthly loyalty plans advertised at stampli.ae.	Wallet stamps are already a low-cost local product. The advertised 2x lift is not used as measured evidence. [M15]
Loopy Loyalty	US$25, US$69 and US$95 monthly plans; feature and staff limits vary.	A managed loyalty pilot can be cheaper than building. Export and unique enrollment are higher-tier features. [M14] [R10]
PassKit	US$39.50 minimum with included pass allowance and volume charges.	Infrastructure buy option, not a substitute for our POS and pickup work. Use contractual rate card. [M13]
Pagers, order-number screens and calling names	Existing operational substitutes; no comparable verified local price established.	They work without a phone, data consent or QR scan. Retain an accessible fallback.

A viable position is: customer-facing pickup visibility and optional rewards, added to a supported POS with little staff effort. Bilingual operation, truthful status, reliable fallbacks and transparent cost can support that position. None is a durable moat alone. Defensibility would develop through connector reliability, distribution partnerships, documented operating outcomes and low onboarding cost.

Market sizing method

Build a location registry from current licence activity codes, then deduplicate entities and branches. Qualify operating model, district, POS/version, pickup volume, current notifier and owner access through a documented sample. Do not apply an arbitrary cafe percentage to all food establishments.

TAM is active locations meeting the operational definition. SAM is the subset reachable with supported integrations, language, geography and service capacity. SOM is constrained by observed qualified meetings per seller-month, paid-pilot conversion, activation, churn and support capacity. An illustrative funnel can test staffing assumptions, but it is not a measured market size. Record denominator provenance and nonresponse bias before an investment claim.

Egypt remains a separate option, not a free extension of UAE economics. CAPMAS metadata establishes a relevant economic-census category, but does not establish a comparable target-cafe count, POS share or willingness to pay. Currency, payment availability, sender registration, entity requirements and price sensitivity need their own decision gate. [M19]

4 Shared platform and initial module scope

The customer should answer three questions quickly: Is this my order, what is its current state, and where do I collect it? Staff should see the next action without switching between several systems. Owners should see operational intervals with trustworthy timestamps and a reason to continue paying.

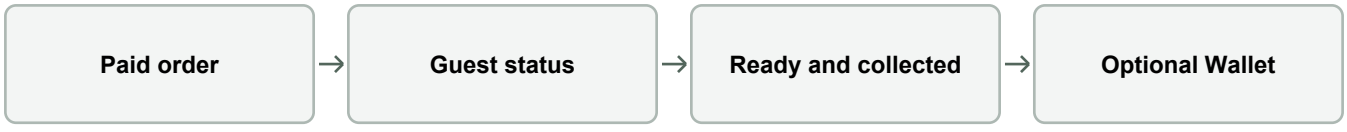
Initial product boundary

Surface	First useful outcome	Release condition
Guest tracking	Open order status from a receipt or ticket, without registration	Opaque order-bound access, no customer PII, expired-link behavior and readable Arabic or English
Staff board	Mark ready and collected quickly; correct mistakes visibly	Location-scoped staff accounts, accessible controls, version conflicts and auditable correction
Manager view	Inspect count, paid-to-ready and ready-to-collected intervals	Sample sizes, missing-data counts, timezone and event-quality indicators
Notifications	One useful ready alert when customer requests it	Recognizable merchant sender, consent, approved route, TTL and spend cap
Wallet loyalty	Save merchant pass and earn from a verified qualifying purchase	Issuer/certificate model, payment truth, claim-once ledger and tested refund policy
Platform operations	Diagnose integration and delivery failures safely	Tenant-safe support access, audit log, runbooks and recovery evidence

Keep payments, menu ordering, inventory, delivery dispatch, table reservations and a consumer marketplace outside the initial product. Import payment facts; do not become a payment processor or POS replacement. Future options have explicit triggers in the innovation portfolio.

5 Order handoff customer and staff journey

Order to return visit



Order tracking remains available when loyalty or notifications are declined.

1. POS completes or records the sale. Our adapter normalizes the order identifier, location and authoritative payment facts. If this path is unavailable, a short pilot uses a separate opaque QR ticket linked by staff; its extra effort is measured.

2. Customer receives a clearly labelled tracking QR, distinct from any unrelated receipt code. The guest page shows merchant, short display number, collection location, current state and last update. It never asks for a phone number just to show status.
3. Customer optionally selects an available notification channel. The page names the cafe, purpose and contact being used. Promotional permission is separate and off by default. Nonusers keep the screen, display or manual-call fallback.
4. Staff or a proven KDS event updates readiness. The page reflects committed server state. A delayed connection says the update is stale; it does not show an invented live countdown or a false ready state.
5. A staff-confirmed collection event closes the operational loop. Notification delivery, reading, opening the page and arriving nearby do not count as collection.
6. After purchase, an optional invitation offers a Wallet card with a clear reward. A secure customer association links future qualifying purchases. The tracking page remains useful to a person who declines loyalty.
7. A qualifying paid order can earn once under the snapshotted program rules. Refunds reverse the appropriate earn through ledger entries. Reprinting or rescanning the receipt never earns again.

For mixed food-and-drink orders, v1 can display only full-order readiness if the merchant agrees to that definition. If partial pickup is essential, model fulfillment units and derive parent readiness. Do not label an order ready because the first item is complete. Reopening a ready order needs a reason and invalidates pending stale alerts.

6 Integration choices and the first technical gate

The receipt QR is not merely a graphic. An order must exist in our system, the link must resolve to that order safely, and the POS or ticket process must place the right link in the customer's hand. Prove this chain on the actual hardware before broad UI work.

Path	Advantage	Main risk and decision
Existing notifier or managed queue tool	Cheapest operational proof; minimal code	May already solve the problem. Buy or partner if it delivers the needed outcome.
Separate opaque QR ticket or sticker	Does not require native receipt modification	Extra handling, printer or ticket stock. Accept only as an instrumented pilot unless staff effort is demonstrably low.
Generic counter QR plus order code	Avoids dynamic print injection	Numeric codes alone are guessable. Use a private pickup secret or pairing step; minimize public data and prevent enumeration.
Read-only POS adapter with staff ready and collected	Sales originate once; no payment rewrite	Event coverage, paid/refund semantics, sandbox access, licensing and reconciliation must be verified.
Native receipt and KDS integration	Best potential end-to-end workflow	Custom URL/QR rights, kitchen events and vendor approval are unproven until tested.
Printer bridge or local agent	May support otherwise closed hardware	Device installation, network permissions, offline jobs and support burden; defer unless customer demand pays for it.

Path	Advantage	Main risk and decision
New POS or online ordering system	Full control	Material category expansion and payment/operations scope. Separate product decision.

Foodics is the first discovery candidate, not a committed partner. Its digital-receipt guide documents OAuth and customer-order created/updated events, using closed and returned sale statuses for receipt logic. That does not prove all anonymous walk-in orders are covered, that native printed receipts accept our QR, or that kitchen readiness is included. Separate documentation says an unlisted integration requires a merchant request and Advanced plan or API licence. [M07] [M16]

The go decision requires sandbox traces for anonymous and identified sales, split payment, void, partial/full refund, reopen, reprint and offline resynchronization. Establish source identifiers, version or sequence behavior, signature verification, replay limits, pagination and rate limits. Obtain written commercial and production-access terms. The supplied integration questionnaire turns these unknowns into procurement questions.

Printed QR proof should use the actual 58 mm or 80 mm printer and receipt layout. Proposed acceptance: at least 95% of representative printed codes open the correct order within ten seconds on tested phones, with no wrong-order matches. Include faded paper, reprints, poor lighting and Arabic text. This threshold is a product target, not a measured result.

7 Notifications and Wallet constraints

WhatsApp

Use the merchant's recognizable sender and merchant-owned assets where supported. Keep business onboarding, number verification, permissions, app review and coexistence as external dependencies. A QR web session does not send a WhatsApp user message and therefore does not open its 24-hour customer-service window. Capture permission for order updates; do not append a sales offer to a transactional ready template. [P03] [P04] [P05]

Meta's current developer documentation reflects pricing effective 1 October 2026. Service messages are charged after the applicable monthly free service tier; utility templates inside the customer-service window are no longer universally free. The free tier is 1,000 delivered service messages per business phone number per month, not a free utility allowance. Older Business-site wording conflicts with these updates. [P01] [P02] [A03]

The archived official AED rate artifact effective 1 October 2026 lists UAE utility and service at AED0.0576 per delivered message and marketing at AED0.2115. These are Meta list rates before tax, provider costs and applicable tiers; recipient market is determined by phone-country rules. A UAE cafe can message tourists whose destination markets differ. Preserve rate-card version and model recipient mix. [A04] [A05]

Send one ready alert by default. Accepted, delivered, read and collected are distinct events. Suppress intents after cancellation, collection, stale versions or TTL expiry. A provider timeout can leave delivery outcome unknown: do not blindly resend and claim exactly-once delivery. Reconcile provider IDs/webhooks where available, then apply a documented bounded retry/fallback policy.

Offer prepaid or capped usage; show estimated and actual spend. Default to one chosen channel. An SMS fallback should require a defined failure condition and customer preference so it does not silently double both messages and cost.

SMS

UAE sender registration and carrier routes need validation before SMS is advertised. Twilio's UAE guidance requires registered alphanumeric sender IDs and describes one-way constraints; its help article currently says new registrations are on hold. This is a Twilio provisioning constraint, not a universal UAE SMS prohibition. Its international sender setup and recurring fees are distinct from message transport. Obtain a written route and sender quote, including Arabic encoding/segments, URLs, recipient coverage, receipts, opt-outs and provisioning time. [P11] [P13] [P14] [A06] [A07]

Marketing SMS rules and transactional ready alerts are not interchangeable. Keep separate purpose, consent and sender classifications. If no permitted economical SMS route exists, disable it and retain web status, on-premise display and staff call-out. SMS availability is not a mandatory launch dependency.

Apple and Google Wallet

Wallet is a loyalty surface, not our authoritative order-alert channel. Apple permits web pass distribution without a companion app; APNs signals an update and the device fetches a newly signed pass. Visible alerts are separate. Apple's HIG says change messages are for time-critical information and must not be marketing or other noncritical communication. [P07] [P08] [E05]

Apple's current licence also restricts Pass Type ID use and signing third-party passes. Proposed lowest-risk model is merchant/brand-owned membership, Pass Type ID and certificate with a contracted service provider. A platform-owned co-branded program requires an explicit contract and issuer-representation review; visual branding alone is not clearance. US$99 yearly membership may apply per issuer rather than once for every merchant combined. Resolve the model before setting Wallet COGS or promising seamless migration. [A01] [A02] [M17]

Google supports loyalty objects and controlled notification APIs, but production publishing requires review. Notification operations have documented limits and user settings. Nearby reminders require precise always-on location and notifications granted to Wallet; Google controls the trigger and wording. Providing store coordinates does not create a customer-location feed. Both platforms need real-device issuance, update, expiry and permission tests. [M18] [P10] [P15] [R12]

8 Brand and experience system

Working brand remains Marrah, with Arabic mark reserved for language and trademark review. No domain, trademark or handle is represented as available. The master descriptor should explain service progress and loyalty for local businesses, with separate order-pickup, service-queue and rewards descriptions. Avoid promising a complete restaurant, retail, appointment or membership operating system.

Proposed platform message: Know what comes next. Give customers a reason to return. Order pickup, queue status and loyalty get sector-specific descriptors rather than calling the whole product a cafe system. Supporting copy should describe order visibility, optional alerts and verified rewards without guaranteed revenue or notification delivery. Merchant identity dominates the receipt page and Wallet card. The platform's own endorsement is secondary and transparent.

Use warm ivory, dark ink and one terracotta accent, with readable Arabic and English typography. Status requires text and icon as well as color. Use accessible primitives, keyboard operation, visible focus, labelled forms, reduced motion and announcements that do not repeat on every poll. A 44 CSS pixel touch target is a proposed usability standard; WCAG 2.2 AA is the acceptance target, not a certificate claimed before testing. [R06]

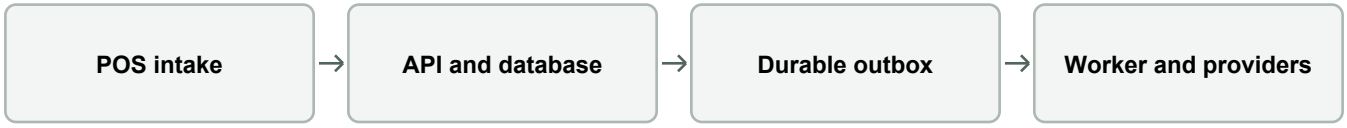
Customer screen inventory: valid tracking, pending/stale connection, partially ready if supported, reopened/problem state, ready with collection instructions, collected/expired link, optional notification

consent, loyalty association, reward balance and Wallet save. Staff inventory: authenticated location selection, live board, order detail, corrections, connection health and exceptions. Managers need staff permissions, POS setup, channel budgets, reward rules, ledger exceptions, analytics and billing.

Require an Arabic/English copy inventory and actual RTL testing. Phone numbers and order codes stay directionally isolated; dates and currency use locale-aware display; server timestamps remain UTC. Customer tests should include a person who declines phone collection, a non-smartphone customer, screen-reader use, large text and poor connectivity. Do not require a customer to subscribe, install or accept marketing to collect food.

9 Architecture and technical decisions

Committed state before external effects



POS payment truth, fulfillment state and notification delivery remain separate.

Recommended baseline is React and TypeScript with Vite for web surfaces, Fastify on Node.js for API commands, PostgreSQL for transactional state and reporting, and a separate worker for integration reconciliation, notifications and Wallet updates. Use a modular monolith: shared domain packages and migrations, separate API and worker processes. This is a proposed fit, not a claim that a particular package version has passed production review.

PostgreSQL owns order projections, audit events, payment facts and reward balances. A transaction commits a domain change and its outbox event together. The worker performs external effects after commit. Separate order fulfillment, payment, reward and notification state machines. A ready drink can later be refunded; a delivered alert is not a collected order.

Server-side events are a suitable first live-update transport, with authenticated or capability-scoped subscriptions and a polling fallback. On reconnection, refetch authoritative state after a gap or version mismatch. Never broadcast an entire tenant's payload to a guest. Customer tokens must not leak through analytics, referrers, logs or shared caches.

Material stack alternatives

Option	When it is reasonable	Decision constraint
React and Node modular monolith	Small TypeScript team; order events and Wallet work	Default if team competence matches. Preserve worker and database boundaries.
Next.js full stack	Existing team strength, marketing SEO or SSR needs	Still needs durable background jobs and explicit authorization.
Laravel	PHP team with stronger delivery experience	Valid alternative with queues and policies; team experience outweighs language fashion.
Edge or serverless platform	Proven regional latency or operating-cost benefit	Validate signing libraries, connection model, long jobs and retry guarantees before replacing conventional Node.
Flutter or native apps	Proven device, offline or printer limitation	Staff-native can be justified before a consumer app; do not add installation for ordinary tracking.
Managed Wallet provider	Faster compliant issuance and certificate operation	Compare pass volume, issuer ownership, residency, export, migration, SLA and support.
UAE managed cloud	Residency, enterprise contracts or measured latency require it	Verify each database, key-management, backup and logging service in the selected region.

Render plus Supabase can support a measured pilot in Frankfurt, subject to transfer review and latency tests. Neither currently lists a UAE region. Their standard plans do not confer an enterprise contractual uptime commitment. A UAE-capable Azure or AWS deployment is a procurement alternative, not an interchangeable price quote. [E01] [E02] [E08] [E09] [E10]

Early published starting components are US$7 monthly for a Render web process, US$7 for a worker and US$25 for one Supabase Pro project. That is a starting configuration, not production capacity or recovery assurance. Supabase PITR is additional paid capability; its cited pricing starts at US$100 per month for a seven-day recovery window. Include backups, monitoring, secrets, logging, network charges, support and operational time when comparing hosts. [R08] [R09]

The technical appendix defines modules, states, entities, API conventions, tests and failure cases. Its proposals are not a completed implementation or an approved vendor contract.

10 Security privacy and reliability baseline

The public receipt is a limited capability, not identity. Generate unpredictable order-bound tokens, store a keyed hash, minimize the guest response, expire or revoke access and rate-limit attempts. Do not expose phone numbers, customer history, payment details or loyalty-account ownership through a receipt. A copied receipt must not permit account merging or repeated rewards.

Derive tenant, location and permission from trusted membership. Every tenant-owned relation and write must preserve that boundary; use composite tenant foreign keys where appropriate. RLS is defense in depth, with separate migration and runtime roles. Never assume a service credential that bypasses RLS is protected by policies. Test cross-tenant reads, writes, exports, streams, queues and support tools explicitly. [R05]

Use secure sessions, MFA for privileged owners/operators, CSRF protection where cookie sessions require it, bounded request bodies, webhook signature/replay verification and secret redaction. Customer

contact verification and recovery must not permit receipt-based account takeover. Support access should be explicit, time-limited and audited.

The merchant normally controls order/customer purposes; platform processing roles, subprocessors, contractual instructions and independent billing purposes must be documented. UAE federal PDPL and DIFC or ADGM regimes have different application and transfer provisions. Choose the applicable regime for the actual entity and flow. Overseas hosting is neither automatically forbidden nor automatically compliant. A UAE region alone also does not cover overseas logs, support access or subprocessors. [P19] [P20] [P21]

Proposed retention defaults need legal and merchant approval: short-lived receipt access after collection, seven-day minimized raw webhook storage, 90-day detailed operational events and longer deidentified aggregates where justified. Loyalty rights, audit evidence, consent proof, deletion requests, legal holds and accounting records need separate rules. Do not set a single deletion clock for every table or retain customer phone numbers merely because storage is cheap.

Proposed public-release objectives are 99.9% tracking availability, connected update visibility within five seconds for 99% of tested events, and p95 staff actions within one second from representative Gulf devices under the agreed load. Provider delivery has its own measurement and cannot be guaranteed by our API SLO. These targets require load and device evidence, a monitoring window and a named owner before customer promises.

A restricted pilot may accept a documented daily-backup recovery risk while preserving the POS as payment truth and manual-call fallback. Public combined release proposes RPO at most five minutes and RTO at most one hour, requiring paid recovery mechanisms and a successful restore/reconciliation drill. Do not claim a five-minute RPO from daily backups. A failed restore or material tenant leak blocks release.

Billing and plan-limit outages must not abruptly strand already accepted orders. Give merchants a clear grace and termination policy while preserving current pickup states. During network or integration failure, show staleness, continue safe read-only displays where possible, queue only operations whose conflicts can be reconciled, and never authorize offline reward redemption against an unverified balance.

11 Measurement and experiments

Use committed order events with source event time and server receipt time. Paid-to-ready includes queue and preparation; active preparation needs a separate start event. Ready-to-collected ends with a staff-confirmed collection. Reopens and partial fulfillment need consistent definitions. Report median and p90 with counts, missing timestamps, exclusions, timezone and source lag; do not average branch percentiles.

Proposed operational pilot: three cafes, two baseline peak-shift observations per location, then two weeks of comparable peak-shift use. Count status questions per 100 eligible orders, valid readiness updates, readiness update lag, ready-to-collected tail, QR-open rate, message opt-in, extra staff seconds per order and support minutes. This is an operational study, not sufficient evidence of lasting loyalty lift.

Initial decision thresholds are hypotheses: at least 30% guest tracking uptake; at least 95% correct readiness updates; at least 25% fewer status questions per 100 comparable orders; no more than five added staff seconds per order; and at least two of three merchants actually paying to continue. Record uncertainty, workload and menu-mix changes. Matched shifts alone do not establish causality; randomize comparable shifts or roll out sequentially where practical and monitor spillover.

For loyalty, freeze a fixed eligible cohort before invitation and randomize within merchant using prior frequency and recency. Analyze all assigned customers, including non-enrollers. Primary metric is incremental contribution after rewards and other costs. Measure earned but unredeemed rewards and

follow customers beyond the promotion to detect purchases pulled forward. Estimate sample needs from observed purchase-margin variance; a small three-shop pilot is not automatically powered. [R04]

Progress framing has relevant research support, including a real cafe reward study and an endowed-progress experiment. These support a testable mechanism, not a guaranteed UAE revenue lift. Test a plain balance against a progress display with the same reward, required paid purchases, expiry and message volume. An honest welcome bonus can be tested while keeping remaining paid requirements equal; record it as bonus progress, never as a fictitious completed purchase. [R02] [R03]

Avoid daily streak pressure for an occasional purchase, public spend leaderboards, gambling-like wheels and rewards for merely scanning receipts. Experiments need an owner, control, outcome, cost guardrail, fraud checks and an end date. Stop an experiment that harms margin, fulfillment or trust even if engagement rises.

12 Business model and financial interpretation

Pricing proposal for validation is AED399 per location per month plus AED799 setup, with a defined order band and separately billed notification usage. This is an assumption, not observed willingness to pay. An AED249 pickup-only offer and an AED599 managed-service offer are alternative price hypotheses to test through actual quotes, not simultaneous launch tiers or observed willingness to pay. Do not bundle unbounded messaging, custom connectors or field visits.

The model separates merchant economics from platform economics. Merchant cash contribution includes incremental purchases, reward fulfillment, displaced paid add-ons, administration, subscription, usage charges and avoided remake waste. Released staff time is shown as capacity value only. It is not cash saving unless hours are actually removed or capacity creates separately measured contribution; counting both would double-count benefit.

Illustrative base inputs are 6,000 paid orders monthly, a fixed eligible-customer cohort accounting for 25% of baseline orders, AED25 ticket, 60% contribution before rewards, 8% purchase lift, and an AED3 upgrade per eight paid program orders. The downside assumes 2% lift, higher notification opt-in and more support. None of these operating inputs comes from a merchant dataset.

The model uses AED0.10 as a landed-channel assumption, not a quote: current UAE Meta utility list rate AED0.0576 plus AED0.0424 unquoted provider/operating reserve. Actual destination mix, provider billing model, tax, sender costs and volume terms can differ. If the merchant pays Meta directly, remove corresponding message cost and resale revenue from our books; do not recognize pass-through revenue twice.

With a 25% message-cost markup, base notification usage is AED187.50 monthly. Base modeled merchant net additional contribution is AED600; downside is a loss of AED950.75. Pickup-only cash benefit under the same assumptions is negative before any measured throughput gain. This is an important falsification result: reduced status questions alone may not support the proposed fee. Willingness to pay and verified operational benefits must be tested rather than manufactured by adjusting inputs.

Platform contribution includes subscription and message usage revenue, actual message cost, payment fees, software reserve and costed support. It separates replacement acquisition in steady-state analysis from actual new-merchant acquisition in the cash ramp. It also shows overhead-only break-even versus merchant count funding the owner-return target. Positive contribution per merchant does not establish sufficient demand, acquisition capacity or funding.

The editable workbook contains Base and Downside controls, shared price and notification assumptions, a 24-month expected-cohort cash ramp and negative-margin guards. Model outputs exclude unquoted

legal/entity costs, unresolved issuer fees unless entered, and any scope not in its editable budget. Growth, churn and initial cash are assumptions. Do not call steady-state recovery at 100 merchants a launch payback forecast.

Financial outputs and funding boundary

Active scenario result in AED	Base	Downside
Merchant net incremental monthly contribution	600.00	-950.75
Merchant pickup-only cash contribution at same full fee	-532.50	-747.00
Additional monthly eligible orders needed to break even	78.97	95.01
Platform sustainable contribution per location	328.39	206.14
Locations funding overhead and owner-return target	40	64
Platform monthly remainder at 100 locations after owner return	19,838.60	7,613.60
Assumed opening cash for the ramp	250,000.00	250,000.00
Minimum modeled closing cash	-76,090.81	-339,324.66
Additional funding gap before safety buffer	76,090.81	339,324.66
Modeled active locations at month 24	58.58	20.95
First positive operating month before build cost	19	Not reached in 24 months

These are model calculations, not market benchmarks. The pickup-only line intentionally removes loyalty gains and reward costs while retaining the full AED399 fee and message usage. It is a diagnostic, not a recommendation to charge that same price for less scope. The separately proposed lower-priced option requires its own recalculation and actual price test.

The 24-month cash example assumes two useful engineers and commercial launch in month 6 for Base or month 9 for Downside, with four or two new paid locations monthly thereafter. Churn, half-month billing for new locations, support and collection costs flow through the same active case. Owner return and fixed overhead start in month 1. Engineering build cost is conservatively front-loaded: 1,160 or 1,868 hours at AED150, equivalent to AED174,000 or AED280,200 for the combined Gate 0 to 4 scope. Provider delays can move launch later. Non-engineering work, entity/legal setup, hardware and unresolved issuer costs are additional; the cash example is not a complete funding quote.

This is the UAE order-handoff plus Wallet worked example, not the funding requirement for every Gulf market or sector. The 24 regional expansion items are excluded. Results use a VAT-exclusive planning basis with the VAT driver set to zero; that does not assert a lawful zero tax rate. Tax registration, applicable place of supply, merchant recoverability, collection/remittance timing and tax-inclusive processor costs require an entity-specific model. Changing the VAT input updates the modeled payment-fee base but does not implement a complete VAT cash-flow ledger.

Use the workbook's minimum-cash and funding-gap outputs to test runway. Positive monthly contribution at a mature merchant count does not mean the initial cash balance survives the ramp. If funding is inadequate, validate with existing tools, narrow the release to pickup only, defer owner draw, improve acquisition productivity, or raise sufficient capital before commissioning the full scope. Recalculate rather

than simply reusing a smaller loyalty-only budget.

Core commercial formulas

Merchant net equals incremental paid-order contribution minus earned reward fulfillment reserve, cannibalized add-on contribution, extra administration, subscription and message usage, plus measured avoided-remake waste. Subscription and messaging are counted once. Released labor is shown separately until monetized.

Platform steady-state contribution equals subscription plus usage revenue minus landed message costs, collection fees, core variable costs, support and churn-weighted replacement CAC. Cash ramp instead charges actual new-location CAC; it does not subtract replacement CAC a second time. Setup contributes only after its direct onboarding and one-time collection costs. One-time setup is not recurring revenue. [R07]

13 Roadmap and release gates

Work begins with proof, then a narrow production path. The backlog contains 75 bounded items with dependencies, owners and acceptance criteria. Effort ranges are aggregate productive worker hours, not elapsed time or binding estimates. Field research, legal review, sales, UX and service operations are separated from engineering. Provider queues and merchant observation periods are additional calendar dependencies and can overlap suitable work.

Workstream	Evidence required to exit	Engineering estimate
Gate 0 market and POS proof	Reachable paid design partners; viable order/QR path; documented access, fees and privacy decisions	48 to 84 hours
Gate 1 custom foundation and instrumented pilot	Authenticated staff, scoped guest access, accurate states, fallback, basic restore evidence and paid field results	248 to 392 hours
Gate 2 one POS pickup MVP	Durable intake, reconciliation, supported QR route, tenant controls and measured pickup acceptance	364 to 592 hours
Gate 3 Wallet loyalty	Approved issuer model, paid eligibility, atomic redemption, refunds, Apple and Google device evidence	276 to 432 hours
Gate 4 messaging and production readiness	Permitted channel, spend control, recovery drill, device/RTL/security evidence and paying beta	224 to 368 hours
Gate 5 expansion	Repeated activation economics, branch permissions and justified second connector	176 to 288 hours
Later experiments	Explicit trigger, control, outcome and cost guardrail	120 to 208 hours if selected

Work in Gates 3 and 4 can run in parallel after its actual dependencies. The separate pickup-only release milestone is G4-013; combined release G4-012 joins it with Wallet acceptance G3-011. Order messaging does not require completed Wallet loyalty. Public pickup-only release can omit loyalty, but must pass the

applicable security, privacy and recovery requirements. Public combined release requires both lanes. SMS may be unavailable; a documented disabled SMS path with permitted alternatives can satisfy channel readiness.

There is a cheaper validation option before custom Gate 1: use an existing tool or concierge workflow with approximately 8 to 24 engineering hours for setup/instrumentation, plus field effort and a two-week observation window. That option tests the problem and payment demand, not production architecture. Choose it when integration access or merchant willingness is uncertain.

The full optional portfolio totals 1,456 to 2,364 engineering hours plus 460 to 786 non-engineering hours. Through custom pickup Gate 2, the engineering range is 660 to 1,068 hours. At an assumed AED150 per engineering hour, that pickup range is AED99,000 to AED160,200 before non-engineering effort, hardware, hosting, legal/entity costs and provider fees. The all-options sum is not a requirement to spend the full amount now.

At 30 productive engineering hours weekly, 660 to 1,068 hours is about 22 to 36 engineer-weeks. Two useful engineers at a combined 60 productive hours weekly imply roughly 11 to 18 working weeks before additional review/field dependencies, not a guaranteed delivery date. Parallel work reduces elapsed time only where dependencies and integration overhead permit it. Re-estimate after Gate 0 and the first working order slice; reuse evidence from already passing checks.

First thirty days

Week 1: qualify 15 to 20 merchants in one district; observe real pickup; collect POS/version and printer details; request sandbox and commercial access; begin messaging and issuer-account decisions in parallel.

Week 2: prove one order-to-link path on actual hardware; freeze event definitions and pilot consent; obtain paid design-partner commitments; choose concierge versus custom foundation. Decline incompatible merchants rather than forcing an unsupported POS promise.

Weeks 3 and 4: run the narrow operational pilot or complete the first custom tracer slice, depending on reuse and access. Measure actual staff effort and data quality. End the month with a recorded go, revise or stop decision, a revised cost range and named unresolved external dependencies. Do not claim a complete production release in thirty days by removing its safeguards.

14 Distribution and operating model

Start with founder-led diagnosis of cafes in a compact district. Show the actual order workflow, not a feature deck. Ask for a paid pilot and a staff champion. Track prospect qualification, owner meeting, access granted, activation, first successful shift, paid conversion and retained location. Include sales time, travel and failed implementations in CAC.

After two repeatable wins, approach local POS integrators, printer dealers and cafe operations consultants. Channel economics need a commission, implementation scope, escalation path and support responsibility. A partner logo is not permission to claim an integration. Enterprise groups follow only after branch controls, support coverage, audit and commercial contracts are ready.

Onboarding should capture branch hours/timezone, supported POS version, order types, display codes, printer path, status authority, merchant sender and approved templates, staff roles, collection instructions and fallback. Train during a quiet period, then observe a peak shift. Record an activation checklist with evidence, not merely a completed form.

Name an on-call owner during advertised supported hours, with POS/vendor escalation contacts and a manual operating plan. Define who fixes bad status, duplicate orders, wrong QR, lost phones, sender

blocks, a failed worker and a refund after redemption. Report weekly support minutes per location and failure causes; support burden is a product metric.

Cancellation/export policy should explain order and loyalty data, reward obligations, certificate ownership, migration limitations and access expiry. Do not hold merchant data hostage to prevent churn. An expiring platform subscription must not silently erase earned customer rights.

15 Innovation portfolio with decision triggers

Explore these options without treating them as committed scope. Score each against merchant profit, staff effort, privacy, integration cost and reversible rollout.

Idea	Useful hypothesis	Trigger and experiment
Calibrated ready-time ranges	Honest uncertainty reduces counter interruptions	Add only after timestamp quality; compare range coverage and questions, not decorative countdown clicks.
Pickup zone or shelf guidance	Clear collection instruction reduces search and wrong pickup	Trial signs and page copy before hardware; measure handoff time and mistakes.
Anonymous repeat-use token	Recognize return use with less form friction	Privacy review and account-recovery design; compare adoption without silently creating marketing profiles.
Off-peak reward eligibility	Existing slack capacity can absorb low-cost rewards	Verify capacity and substitution; controlled contribution-margin experiment.
Honest welcome progress	Same remaining purchases can feel more achievable	Equal reward cost and requirements; bonus clearly labelled, no fake purchase history.
Adaptive notification choice	Customers prefer different channels and have different delivery markets	Explicit preference, price visibility and one-channel default; measure successful pickup and landed cost.
Automatic receipt pairing	Existing customer identifier avoids an extra loyalty scan	Requires reliable POS identity and permission; measure wrong-account and fraud rates.
Delay explanation and staff exception queue	Useful context improves trust and recovery	Define controllable reasons; test complaints and burden, not an unsupported promise of faster cooking.
Workload-based preparation forecast	Station/menu mix predicts readiness better than FIFO	Only after enough clean events; backtest by branch and publish calibration before live ETA.
Merchant experiment dashboard	Owners can distinguish member sales from incremental contribution	Requires holdout design and sufficient sample; show uncertainty and cost allocation.
Native staff companion or printer bridge	Specific device limitations justify installation	Build only after measured browser/printing bottleneck and economic commitment.
Regional rollout and multi-brand rewards	Distribution or customer reuse improves economics	Separate country/legal/PSP study; do not pool reward liability or customer identity by default.

Do not market AI forecasting before a simpler baseline is accurate. No automated dynamic price, hidden profiling or cross-merchant customer graph is required for v1. Native consumer apps, payments, delivery and marketplace discovery need separate evidence and investment decisions.

16 Acceptance and decisions before production

Technical acceptance must demonstrate customer-visible behavior under realistic failure: duplicate and reordered POS events converge; stale staff writes fail safely; one eligible order earns once; concurrent redemptions spend once; refund replay reverses once; wrong-tenant users and streams return no data; expired receipt links reveal nothing; worker restart preserves committed jobs; unknown message outcomes do not create retry storms; reconnect restores truth; recovery rebuilds order and ledger projections.

Operational acceptance requires an observed shift with honest statuses and fallback, understandable Arabic/English instructions, customer choice without forced registration, a signed channel/issuer decision, and a merchant paying a viable price. Public release also needs completed privacy/vendor contracts, a documented recovery result, named service owner, rollback plan, valid certificates, approved production publishing and a costed support promise.

Open decisions needing external evidence are exact launch jurisdiction and invoicing entity; first POS merchant/plan/version; native versus separate QR route; anonymous-order event coverage; staff versus KDS readiness authority; permitted notification provider and current all-in fees; Apple issuer ownership and managed-provider terms; reward/account recovery rules; transfer and retention arrangements; actual purchase margins, uptake, support, CAC and churn.

The next production action is a Gate 0 evidence sprint using the integration questionnaire and backlog. Its output should select or reject one concrete merchant/POS path. All later estimates and provider choices must be revised against that result before irreversible commercial commitments.

17 Technical specification appendix

The following contract and state proposals support implementation planning. They require a selected POS adapter, security review and implementation tests before freezing. Exact Apple and Google vendor protocols remain governed by current official documentation; do not infer them from our internal API conventions.

State models: keep axes separate

Fulfillment state

`received -> preparing -> ready -> collected`

Terminal/exception transitions: `received|preparing|ready -> canceled`; `ready -> preparing` for a documented reopen; `collected -> ready` only through privileged correction, never normal UI. Every transition records actor, reason, source, `occurred_at`, server `recorded_at`, and monotonic order version.

- **Partial orders:** parent order derives display state from fulfillment units/batches. Each unit can progress independently. Parent becomes `ready` only when all non-canceled required units are ready; `partially_ready` is a derived presentation label, not a writable state. Parent becomes `collected` only when every non-canceled unit is collected. UI shows “2 of 3 ready.”
- **Reopened ready:** staff can send `ready -> preparing` with mandatory reason (quality remake, missing item). Increment version, emit `order.reopened`, suppress stale “ready” deliveries via event version, then notify updated status.
- **Corrections:** never rewrite history. Append correction event; project current state. Privileged collected-state correction requires manager role and audit alert.

Payment state

Separate axis: `unknown -> pending -> paid -> partially_refunded -> refunded`; `pending -> failed|voided`. Provider/POS event time may precede receipt. Payment events never move fulfillment automatically except an explicit tenant policy such as “do not start before paid.” A refunded prepared drink may remain collected; reporting must show both facts.

Notification state

`planned -> queued -> submitted -> provider_accepted -> delivered -> read`, with `failed`, `unknown`, `expired` and `suppressed` branches. Provider acceptance is not delivery. Only a trustworthy provider delivery signal records delivered; a timeout without final evidence records unknown. Read is recorded only when supported and evidenced. Wallet APNs updates are hints: Wallet fetches current pass after a push [E05]. Notification/read status never means order collected.

Reward state

Eligibility: `ineligible|pending_payment|eligible -> claimed -> reversed`. Value ledger entries are `earn`, `redeem`, `expire`, `adjust`, `reverse`; balance is a projection, not editable truth. Refund reversal is its own entry linked to original earn.

Integration and failure semantics

POS intake and truth

Define an adapter contract: authenticate webhook, parse provider event, map location/order/payment identifiers, store raw digest, upsert normalized facts, and acknowledge. The POS remains authoritative for receipt totals, tenders, refunds, and canonical sale identifier. This platform owns fulfillment annotations, receipt-access tokens, notifications, and loyalty calculations derived from received facts.

Webhook transaction:

1. Verify signature against tenant/location secret before JSON interpretation where provider supports it.
2. Insert `webhook_inbox(provider, tenant_id, external_event_id, payload_hash)`; unique constraint makes retries idempotent.
3. Compare provider sequence/version when available. If absent, apply field-specific timestamps and never regress terminal facts. Store unmatched/out-of-order events pending their prerequisite.
4. Update normalized records and outbox atomically; return 2xx only after commit. Duplicate with same ID/hash returns 200. Same ID/different hash is quarantined and alerted.
5. Worker retries inbox failures with exponential backoff and jitter. Poison payloads move to review queue; never retry forever.

During downtime, provider retries are preferred. Independent reconciliation polls orders modified since a high-water mark with overlap (for example, last successful cursor minus ten minutes), deduplicates by external ID/version, and records gaps. After recovery, reconciliation must complete before declaring data current. Staff UI shows freshness timestamp and degraded banner. Never fabricate status from a missing webhook.

Receipt QR and printer capability spike

Prototype before committing POS integration:

- **Path A—POS receipt customization:** provider prints a URL QR containing an opaque, 128-bit-or-greater random token. Validate receipt-template variables, QR density/quiet zone, thermal width, encoding, and whether token can be returned synchronously at sale time.
- **Path B—local bridge:** signed, auto-updating store appliance/service watches POS output and sends ESC/POS or vendor SDK print job. High operational burden: LAN discovery, USB permissions, spool retry, duplicate receipts, certificate rotation, remote support. Use only if provider officially permits it.
- **Path C—separate staff/customer display:** staff scans/enters short order code; customer scans a counter/table QR then enters printed pickup code. No receipt integration, but more friction and shoulder-surfing risk. Rate-limit attempts; code alone exposes minimal projection.
- **Path D—preprinted generic QR:** opens tenant page where customer enters order code. Fastest pilot, no unique QR printing.

Test matrix: 58/80 mm thermal printers, iOS/Android cameras, low light, damaged paper, Arabic/Latin content, network loss, reprints, duplicate sale, canceled order, and 30+ minute delayed POS event. Acceptance: >=99% first-scan success over 200 representative prints, no token truncation, reprint maps to same order, and degraded path is documented. Do not put name, phone, email, loyalty balance, order ID, sequential token, or tenant secret in QR.

Receipt access security

Store only a keyed hash of receipt token; compare constant-time. Token maps server-side to tenant/order and expires after configurable retention. Rotate by issuing a new token and revoking old one. Public response contains minimum order alias, item readiness summary, timestamps, and branding—no payment instrument or customer PII. Rate-limit by token/IP/device signals without treating IP as identity. Authorization always scopes lookup by resolved tenant and capability. Knowing an order UUID or receipt number yields nothing. Linking a receipt to an authenticated loyalty account requires step-up proof defined by product/security; receipt possession alone cannot merge accounts or expose history.

Loyalty correctness and abuse controls

Default eligibility: exactly one claim per eligible, paid order; eligibility policy version is snapshotted at earn time. Enforce with `UNIQUE(tenant_id, order_id, reward_program_id, claim_kind)` plus a serializable transaction or row lock. Handler re-reads payment/refund state, inserts claim and ledger entry atomically, then writes outbox event. Concurrent scans yield one success and deterministic `409 reward_already_claimed`; they never double-credit.

Refund processor reverses proportional or whole earn according to snapshotted policy. Each reversal has `UNIQUE(original_ledger_entry_id, source_refund_id)`. Never mutate/deletes original ledger entries. Negative balance policy must be explicit: allow debt, cap at zero while recording unrecovered amount, or flag account; do not silently discard. Corrected/refunded events can arrive before claim: eligibility projection blocks claim; later claim attempt explains ineligibility.

Fraud controls: separate cashier, manager, tenant admin, support, and service roles; manager approval for manual earn/reverse or collected correction; per-staff velocity thresholds; location-bound access; immutable audit log; reason codes; daily exception report; alerts for repeated claims, high adjustments, refunds after claim, and same staff/customer concentration. Support impersonation is forbidden; time-boxed audited support grants instead. Fraud scoring may queue review but must not silently remove earned value.

Tenancy, roles, secrets, certificates

Every tenant-owned row includes `tenant_id`; location-owned rows also include `location_id`. API derives tenant/membership from authenticated session, never accepts tenant selection in body without verifying membership. Repository methods require a tenant context. PostgreSQL RLS is defense in depth: enable on exposed tables, force RLS where applicable, test cross-tenant denial, and keep worker/service roles narrowly scoped. Supabase states frontend Data API access requires RLS and no-policy RLS denies requests [E04], but this proposal still routes business commands through API.

Roles: platform operator (no default data access), tenant owner, tenant admin, location manager, staff, analyst/read-only, integration service. Permissions are capabilities (`order.transition`, `reward.adjust`, `billing.manage`, `certificate.rotate`) plus location scope—not role-name conditionals scattered across handlers.

Tenant billing account and subscription state are separate from café payment facts. Plan limits are enforced server-side with soft warnings then defined hard limits; billing outage must not stop order status. Integration secrets live in managed secret/KMS storage, referenced by version ID; database stores metadata only. Encrypt provider tokens, rotate without downtime, redact logs, and audit reveal/replace actions.

Wallet Pass Type certificates/private keys require strict custody. Apple requires an Apple-issued Pass Type ID certificate for recognized passes; expiry prevents new signing and updates, while revocation can break passes [E06]. Use KMS/HSM or encrypted secret store, worker-only access, separate dev/prod

identities, dual-control rotation, 60/30/14-day expiry alerts, signed artifact tests, and a rehearsed rollover. Default proposal is merchant-controlled developer membership and Pass Type ID, with an Apple-issued certificate under that developer team with the platform acting under a compliant service-provider contract. Do not sign unrelated third-party merchant passes using a shared platform ID as an MVP shortcut. A genuinely platform-owned co-branded program needs explicit Apple-contract and counsel clearance before build; account/issuer cost and migration rights must be priced. [A01]

Data model and indexes

Core relationships (UUID/ULID internal keys; all timestamps UTC):

- `tenants 1-* locations`; `users *-* tenants` through `memberships(role, status)` and `membership_locations`.
- `orders(tenant_id, location_id, external_order_id, display_code, source, version, occurred_at)` 1-* `order_lines`; 1-* `fulfillment_units`; 1-* append-only `order_events`.
- `payments(order_id, external_payment_id, state, amount, currency, provider_version)` 1-* `refunds`.
- `receipt_capabilities(order_id, token_hash, expires_at, revoked_at)`; never store plaintext token.
- `reward_accounts(tenant_id, customer_subject_id)` 1-* `reward_claims(order_id, policy_version)` 1-* `reward_ledger`; projection `reward_balances` rebuildable from ledger.
- `wallet_passes(tenant_id, reward_account_id, pass_type_id, serial, auth_token_hash, current_version)` 1-* `wallet_registrations(device_library_id_hash, push_token_encrypted)` and `wallet_pass_versions`.
- `webhook_inbox`, `integration_cursors`, `outbox_events`, `notification_intents`, `delivery_attempts`, `audit_events`.

Required constraints/indexes:

- unique `(tenant_id, source, external_order_id)`, `(tenant_id, provider, external_event_id)`, `(pass_type_id, serial)`, and reward keys above;
- B-tree `(tenant_id, location_id, fulfillment_state, updated_at DESC, id)` for live board; `(tenant_id, occurred_at DESC, id)` for history; `(order_id, recorded_at, id)` for event replay;
- partial outbox index `(available_at, id) WHERE published_at IS NULL`; inbox retry `(next_attempt_at) WHERE status='retry'`;
- partial capability index on `token_hash WHERE revoked_at IS NULL`; notification dedupe unique `(tenant_id, channel, dedupe_key)`;
- BRIN on high-volume append-only event timestamps after volume proves useful; avoid speculative indexes. Query plans and write amplification decide.

Partition events by time only after measured need. Tenant partitioning complicates migrations and uneven tenants; do not start there. Define retention: raw webhook payload days, audit years, receipt token days, operational events/reporting aggregates per legal/product review.

HTTP/API contract

Base `/api/v1`; JSON success `{data, meta?}` and error `{error:{code,message,details?,request_id}}`. UUIDs opaque. Cursor pagination for event/history lists. Mutation commands require `Idempotency-Key` (tenant + route scoped, request hash retained); replay returns original status/body, conflicting body returns `409 idempotency_conflict`. `If-Match: "<version>"` prevents stale staff writes. `401` missing/invalid auth; `403` known principal lacks permission; tenant-scoped unknown/inaccessible resource returns non-enumerating `404`; `409` state/body-version/idempotency conflict; `412` failed If-Match precondition; `422` semantic validation; `429` with `Retry-After`; `503` for temporary dependency failure.

Method/path	Auth	Purpose / invariants
`POST /integrations/v1/pos/{provider}/webhooks`	Provider signature	Durable idempotent intake; 2xx only after inbox commit. Tenant resolved from endpoint key, not payload.
`POST /api/v1/integration-reconciliations`	tenant admin/service	Request bounded location/time reconcile; `202`; one active scope job.
`GET /api/v1/orders`	staff + location	Cursor list/filter; tenant/location scoped; stable `(updated_at,id)` cursor.
`GET /api/v1/orders/{orderId}`	staff + location	Internal full projection; `404` across boundary.
`POST /api/v1/orders/{orderId}/fulfillment-transitions`	staff/manager	Command with expected version, target, reason; validates transition and partial-unit rules.
`POST /api/v1/orders/{orderId}/corrections`	manager	Append correction, never overwrite audit/history.
`GET /r/{receiptToken}`	public capability	Minimal receipt projection; token hash lookup, expiry/rate limit, no account identity. HTML shell may call next endpoint.
`GET /api/v1/receipt-orders/{receiptToken}`	public capability	Minimal JSON; `404` for invalid/expired; cache private/no-store.
`POST /api/v1/orders/{orderId}/reward-claims`	customer session + receipt proof	One eligible paid claim; atomic constraint; `409 already_claimed`, `422 not_eligible/refunded`.
`GET /api/v1/reward-accounts/{accountId}`	owning customer or scoped staff	Balance + cursor ledger; never infer access from receipt.
`POST /api/v1/reward-accounts/{accountId}/adjustments`	manager	Bounded adjustment, reason required, velocity control/audit.
`GET /api/v1/wallet-passes/{passId}.pkpass`	pass capability/customer	Generates/serves signed immutable version; no private key leaves worker boundary.
`GET /api/v1/order-stream`	staff/customer auth/capability	SSE tenant/order-scoped event summaries, resumes via `Last-Event-ID`; no sensitive payload.

Rate limits differ by trust: aggressive for receipt guesses/claims; per-tenant quotas for staff/API; signature endpoints protected by body-size, concurrency, and replay limits. The accompanying OpenAPI proposal is the initial machine-readable contract. Freeze it only after adapter discovery, security review and contract tests.

Real-time, outbox, and notification economics

API transaction writes domain change and `outbox_events` together. Workers claim with `FOR UPDATE SKIP LOCKED`, publish idempotently, and mark success. Consumers dedupe on event ID. Outbox lag and oldest age are primary health signals. A periodic sweeper recovers abandoned locks.

SSE is default: server emits versioned `order.updated` hints, heartbeat every ~20–30 seconds, supports `Last-Event-ID`, and closes/re-authenticates before credential expiry. Clients always refetch authoritative projection after gap/version mismatch. If SSE fails, use exponential reconnect with jitter then polling. A closed browser tab does not receive an automatic web alert merely because the customer scanned a QR; web push has device/install/permission conditions and cannot replace the selected messaging route. [R01] Polling proposals: active order 5–10 seconds, ready order 15–30 seconds, background tab 60 seconds; stop at terminal/expiry. Avoid broadcasting full order data.

Notification intent key (`tenant, order, channel, event_type, order_version`) suppresses duplicates. Coalesce rapid changes; a reopened order supersedes stale ready intent before send. TTL examples: ready alert 30 minutes or order collection, pass-balance update 24 hours. APNs push is a change signal, not payload truth [E05]. Cap retry count/age, honor permanent token failures, batch Wallet updates, and expose per-tenant channel budgets. SMS is opt-in and costly: only meaningful transitions, tenant spend cap, no retry storm. Delivery/read never changes fulfillment.

Apple requires server-side re-signing for changed passes because device-side change invalidates signature [E05]. Node worker handles deterministic manifest/package build and signing. Keep generated pass version immutable and content-addressed. Managed PassKit remains an interchangeable adapter.

Analytics definitions and clock quality

Use event-time definitions with both `occurred_at` (source claim) and `recorded_at` (our clock). Report source and ingestion lag; exclude/flag records outside quality bounds.

- **Receive-to-ready:** first valid `ready.occurred_at` minus normalized `received.occurred_at`; reopened orders also report final-ready and remake count.
- **Ready-to-collected:** first valid collected minus final preceding ready. Never use notification delivery.
- **Paid-to-ready:** ready minus paid for orders with both facts, segmented by source/location/channel.
- **Claim rate:** distinct eligible paid orders claimed / distinct eligible paid orders whose claim window closed in cohort. Avoid immature denominator.
- **Refund-after-claim rate:** claimed orders later partially/fully refunded / claimed orders.
- **Notification metrics:** provider acceptance / attempted, evidenced delivery / submitted, failure / submitted and unknown / submitted are separate rates by channel/provider. Read is separate where supported. Report missing callbacks and unresolved attempts; never count provider acceptance as delivery.

Publish median and p90 only with sample count, window, timezone, and exclusion count. Percentiles are computed from order-level durations, not averages of location percentiles. Monitor POS event inversion, clock offsets, missing timestamps, duplicates, and ingestion lag. If source clock confidence is low, use recorded-time operational metrics and label them; do not mix silently.

Reliability and recovery targets

Proposed application SLOs for paid pilot, measured at external probes and excluding only documented maintenance: API read availability 99.9%; accepted command availability 99.5%; p95 staff read <500 ms and public receipt read <700 ms in host region; 99% webhook commits <2 s; 99% committed outbox events begin processing <60 s; 99% SSE-visible updates <5 s when connected. These are internal objectives, not provider promises.

Data targets: **RPO <=5 minutes** for operational database, **RTO <=60 minutes** for single-region restore, and reconciliation completion <=30 minutes after service recovery for pilot volume. Required mechanisms: PITR/backups, encrypted off-provider logical export or tested independent backup, migration manifests, secret recovery, and documented DNS/deploy restore. Supabase notes PITR can restore to a chosen point with second-level granularity, but plan/configuration must be verified [E11]. Render paid database PITR retention and HA vary by plan [E09].

Test quarterly: restore latest backup into isolated project; checksum tenant/order/ledger counts; replay outbox safely; rotate receipt/pass secrets; rebuild reward balance; simulate worker death, duplicate/out-of-order webhooks, database failover, provider outage, APNs rejection, expired pass certificate, and POS reconciliation. Record measured RPO/RTO. Monthly error-budget review freezes risky releases after burn. Because standard Render/Supabase plans lack contractual uptime SLA [E09][E10], either accept pilot risk, buy Enterprise commitments, or use a provider architecture whose contract supports customer commitments.

Screen inventory and accessibility/RTL

Customer: QR landing, live order tracker with partial items, problem/reopen message, reward claim/sign-in, reward balance/history, Add to Apple Wallet, privacy/preferences, expired/invalid receipt. Staff: sign-in/MFA, location chooser, live board, order detail/transition/correction, exception/reconciliation queue. Manager: staff/roles, locations/POS, reward policy, manual adjustment review, notification templates/budgets, Wallet certificate health, analytics, audit. Platform ops: tenant provisioning/billing metadata, health/support grants—no broad customer data browser by default.

Meet WCAG 2.2 AA target: semantic headings/landmarks, keyboard-complete board and dialogs, visible focus, 44px targets, text contrast, error summaries, no color-only state, reduced motion, live-region announcements that do not spam on polling, and explicit confirmation for irreversible commands. Status chips pair icon/text. Test 200% zoom and screen readers. Localize strings, plurals, dates/numbers/currency; set `dir` by locale; use CSS logical properties; mirror directional icons only when semantic; keep order codes/phone numbers appropriately isolated with bidi markup. Arabic receipts and pass fields need device tests, not screenshot confidence.

Observability, verification matrix, and delivery sequence

Structured logs: request/event/tenant/location IDs (hashed or internal), route, latency, outcome, version—never receipt/pass tokens, payloads, PII, secrets, or push tokens. Metrics: HTTP RED, DB saturation/locks, inbox duplicates/quarantine/lag, reconciliation gaps, outbox age/retries, SSE connections/gaps, notification outcomes/cost, pass signing latency/failures/certificate days remaining, reward conflicts/reversals/adjustments. Distributed traces across request -> DB/outbox -> worker/provider with sampling. Tenant-safe audit export. Alerts use user-impact symptoms and runbooks.

Test layers: domain transition/property tests; ledger invariant and concurrent-claim tests; tenant/IDOR/RLS tests; webhook signature/idempotency/permutation fuzzing; migration/backfill tests; API contract tests; POS sandbox plus recorded fixtures; Wallet package validation/on-device install/update/unregister; printer scan study; accessibility automation plus keyboard/screen-reader manual pass; load/soak for peak orders, SSE fan-out, worker backlog; chaos/recovery drills.

Device/POS matrix: current and previous iOS Safari/Wallet, Android Chrome and Google Wallet (tracking, approved loyalty issuance, save, balance update, permissions and invalid-object handling), desktop Chrome/Edge/Safari for staff, poor 3G/Wi-Fi handoff/background-resume, 58/80 mm printers, each POS provider/version/location topology, refund/void/reopen/reprint/offline sync, English/Arabic and RTL, camera permission denied, expired QR/pass certificate.

Tracer-bullet plan

1. **Domain skeleton and threat model** (dependency: product decisions on POS, claim policy, retention). One tenant/location, migrations, order/payment/fulfillment events, audit. Acceptance: forbidden transitions and cross-tenant reads fail; event replay produces same projection.
2. **One POS fixture adapter + inbox/reconcile** (1). Acceptance: duplicate and every tested ordering permutation converge; downtime drill catches up with no double events; freshness visible.
3. **Staff board + public receipt capability** (1–2). Acceptance: opaque token, no IDOR/PII, SSE plus polling recovery, keyboard/RTL baseline, partial/reopen/correction paths demonstrated.
4. **Reward ledger slice** (1–3). Acceptance: 100 concurrent claims yield one earn; partial/full refunds create exact idempotent reversals; balance rebuild matches projection; fraud audit emitted.
5. **Notification outbox** (2–4). Acceptance: commit survives worker crash, stale ready alert suppressed after reopen, TTL/duplicate/budget rules verified, backlog dashboard/runbook exists.
6. **Wallet build-vs-buy spike** (4–5). Acceptance: real device installs pass, balance update fetched after APNs/pull refresh, expired/revoked-token behaviors tested; managed vendor scored on cost/residency/export/SLA. No certificate in DB/log.
7. **Receipt printer capability spike in parallel with 2–3**. Acceptance: chosen path passes scan/reprint/offline matrix; if no integration path passes, adopt generic QR + code and update UX/cost plan.
8. **Production hardening** (all). Acceptance: backup restore and region rebuild meet measured targets; load/SLO test, accessibility/manual device matrix, secret/certificate rotation, incident runbooks, retention jobs, dashboards, and security review complete.

Do not build multi-POS abstractions, microservices, native apps, advanced segmentation, or multi-region active/active before tracer bullets prove the operational model. First production gate requires named on-call owner, POS escalation path, privacy/legal review, Apple account/certificate owner, rollback procedure, and pilot limits.

Additional implementation invariants

Use a composite tenant foreign key, not merely a tenant column, when linking orders, claims, accounts and locations. Runtime database roles must not own tables or possess BYPASSRLS; transaction-local verified tenant context must not leak through pooled connections. Background jobs carry an authorized tenant context and do not trust tenant fields in raw provider payloads.

Display order numbers can repeat by location or day and must never be global identity keys. Link identity to provider stable order ID plus tenant/location namespace. A public GET, including a messaging link-preview fetch, never claims loyalty, redeems a reward or consumes a one-time pairing credential. Claims and redemptions use explicit authenticated or step-up commands.

Preserve token secrecy through web server logging configuration, Referrer-Policy, private/no-store responses, no third-party analytics on receipt pages and a strict content policy. Hashing a database token does not stop leakage from a logged URL. Staff must confirm collection; delivery receipts and customer page views are not proxies.

For partial refunds, define stamp reversal against qualifying spend or an order rule explicitly. If a spent reward cannot be recovered, record a reviewable adjustment or account debt according to disclosed terms; do not silently rewrite history or delete the customer's other earned value.

Our proposed OpenAPI file defines the core business surface only. It has no claim of running functionality. POS webhook signature names, rate limits, source versions, receipt-print fields, OAuth options and Apple protocol details require the chosen provider's actual integration contract. Structural JSON/reference validation is not interoperability or security evidence.

18 Evidence register

Research access date is 27 October 2026 unless a source entry states otherwise. First-party commercial statements establish advertised terms, not independently verified outcomes. Archived rate files include their effective date and SHA-256 provenance.

M01 Annual Dubai Gastronomy Industry Report 2024

Dubai Department of Economy and Tourism. Almost 1,200 new restaurant licences in 2024; 1,100+ diner survey; 2.5 restaurant visits/week; 18.72m overnight visitors; accommodation/food services +3.7% in first nine months of 2024 and 3.4% of GDP.

Limit: Restaurant licences are not active café locations; survey is consumer sentiment and DET methodology is only summarized on page.

Open source

M02 Over 34,000 food inspections conducted in first half of 2025

Dubai Municipality. 34,700 inspections and approximately 2,336 new food establishments in H1 2025.

Limit: Food-establishment scope is broader than cafés/restaurants; openings not net active stock.

Open source

M03 Over 26,000 food establishments operate Dubai

Emirates News Agency quoting Dubai Municipality. Dubai Municipality official stated more than 26,000 food establishments operate in Dubai.

Limit: Broad food-establishment definition; no activity-code breakdown or deduplication method.

Open source

M04 1.1 million food products registered in Dubai Municipality Food Import and Export System

Dubai Municipality. More than 23,400 food establishments operated in Dubai at publication.

Limit: Older, broad category; not comparable directly to restaurant licences.

Open source

M05 Small to Medium Sized Enterprises Market Conduct Regulation

Central Bank of the UAE. Federal SME thresholds: micro service <5 employees or <AED2m revenue; small service 6–50 employees or <AED20m revenue.

Limit: Definition for regulation/segmentation, not count of SMEs or cafés.

Open source

M06 Foodics Pricing UAE

Foodics. Public UAE monthly prices billed annually: AED199 Starter, AED375 Basic, AED556 Advanced; page mentions digital receipts, loyalty, BI and API integration.

Limit: Page feature layout is ambiguous; hardware, implementation and exact entitlement require quote.

Open source

M07 Integration Availability

Foodics Help Center. Unlisted integration is merchant-requested; merchant needs Advanced plan or API licence; Foodics says it supports the technical team.

Limit: May change; no API licence price or approval SLA.

Open source

M08 Integrate with Simphony Point of Sale

Oracle. Simphony offers Transaction Services Gen2, BI API and JavaScript extensibility; partners enrol in OPN and submit integration inquiry.

Limit: No UAE pricing or proof of custom receipt-QR permission.

Open source

M09 Syrve UAE Pricing

Syrve. AED250/350/450 per till/month ex VAT, annual AED2,500/3,500/4,500; page lists SMS, loyalty, customer display, analytics and APIs by tier.

Limit: Entitlements, setup/hardware and API partner access need confirmation; vendor claims not independent.

Open source

M10 Unified Restaurant Operations Platform and Integrations

Grubtech. Offers order management, KDS, delivery and analytics; integration directory lists POS connections including Foodics.

Limit: No public price; no receipt-template rights shown.

Open source

M11 Deliverect Pricing

Deliverect. Restaurants package includes POS integration, order/menu tools and analytics; pricing is quote-based on location, scale, locations and solution.

Limit: No UAE amount; feature availability and implementation need quote.

Open source

M12 Queberry Queue Management

Queberry. Offers queue management, WhatsApp queueing, digital signage and business analytics, serving UAE and regional markets.

Limit: No public pricing or café-specific receipt integration evidence.

Open source

M13 PassKit Pricing Rate Card

PassKit. Starts USD39.50/month; includes up to 250 multi-use and 250 single-use passes; tiered usage pricing.

Limit: Taxes, team size, overage and enterprise/support requirements may change total.

Open source

M14 Loopy Loyalty Pricing

Loopy Loyalty. USD25/69/95 monthly for 1/3/10 locations, Apple/Google Wallet support and unlimited customers; annual USD250/690/950.

Limit: Fair-use and product-fit constraints; not integrated pickup tracking evidence.

Open source

M15 Stampli – Digital Loyalty Cards for Apple & Google Wallet

Stampli UAE. Advertises UAE digital stamp cards at AED69/179/369 monthly for Starter/Growth/Pro; Apple and Google Wallet passes, push notifications and staff scan tool; Pro includes multi-location support.

Limit: The same page claims a 2x typical repeat-visit lift, customer counts and testimonials without disclosed methodology, sample, comparison period or independent corroboration; these claims are not used as evidence.

Open source

M16 Digital Receipt Integration

Foodics Developers. Documents OAuth, account/receipt settings, customer-order created/updated webhooks, and closed/returned receipt-status logic.

Limit: Does not prove custom QR injection into native printed receipt, all-order coverage, or kitchen ready event.

Open source

M17 Choosing a Membership

Apple Developer. Apple Developer Program listed at USD99 per membership year; advanced capabilities require membership.

Limit: Local currency/tax and exact Wallet certificate workflow must be checked at implementation.

Open source

M18 Google Wallet Loyalty Object REST Resource

Google for Developers. Loyalty objects support account identifiers, loyalty points, barcodes, messages and lifecycle state.

Limit: No authoritative price statement found in reviewed source; approval and operational constraints remain.

Open source

M19 Egyptian Economic Census 2022/2023

CAPMAS. Two-digit activity/governorate publication; sample universe 471,000 establishments; 96.68% response; questionnaire includes restaurants, food stores, cafés and drinking activities.

Limit: Combined category; metadata page supplies no target-café total or POS/digital adoption measure.

Open source

P01 Pricing on the WhatsApp Business Platform

Meta for Developers. Per-delivered-message categories and rates; service charged from 2026-10-01 after 1,000 monthly free messages per business phone number; in-window utility charged; current rate-card links and update calendar.

Limit: Linked CSV/PDF CDN URLs are mutable; capture effective-date artifact during commercial sign-off.

Open source

P02 Upcoming pricing updates for Meta Business Agent, service and utility messages

Meta for Developers. Effective 2026-10-01 service and in-window utility become per-message charged; service rate equals utility/authentication market rate; payment method needed after free tier.

Limit: AI Provider exception requires separate classification; ordinary cafe automation treated as standard third-party tooling, not AI Provider.

Open source

P03 Template categorization

Meta for Developers. Template category determined by message objective/content; promotional or mixed intent is marketing, transactional order information can be utility.

Limit: Final category and approval remain Meta review outcomes.

Open source

P04 Get opt-in for WhatsApp

Meta for Developers. Need phone number plus opt-in for named business; website/SMS/IVR/paper/in-person methods allowed; category-specific expectations/opt-outs recommended.

Limit: Local law can impose stricter consent requirements.

Open source

P05 Embedded Signup

Meta for Developers. v4 onboarding, merchant asset ownership, permissions/App Review, 10-to-200 onboarding limits, payment/credit-line responsibilities, token/webhook steps.

Limit: Account-model evolution marked beta and subject to change.

Open source

P06 Onboard WhatsApp Business app users (Coexistence)

Meta for Developers. Existing Business App number can coexist with Cloud API; app remains usable one-to-one and history syncs; customized onboarding and phone verification required.

Limit: Merchant eligibility and rollout may vary; validate in live onboarding.

Open source

P07 Getting Started with Apple Wallet

Apple Developer. Pass distribution/update model; expired certificate blocks new signing/updates while installed passes continue; revocation can break passes.

Limit: Program fees and supported OS/device matrix may change.

Open source

P08 Wallet Human Interface Guidelines

Apple Developer. Use change messages only for time-critical information; never for marketing/noncritical communication.

Limit: Design guidance is normative platform guidance but enforcement details are not published.

Open source

P09 Apple Developer Program License Agreement — Additional Terms for Passes

Apple Developer. Pass Type ID cannot be shared/transferred except limited service-provider use; cannot sign third-party pass under own ID; pass-related promotional APNs must satisfy terms/consent.

Limit: Contract interpretation and merchant/platform branding arrangement need counsel.

Open source

P10 Trigger Push Notifications — Generic pass

Google for Developers. TEXT_AND_NOTIFY max three per pass/24h; possible throttling; notifications user-controlled; nearby requires precise always-on location; 10 locations per class and object; Google controls trigger/text.

Limit: Feature behavior and permissions vary by device/account.

Open source

P11 Cellular Phone Spam Regulatory Policy

UAE TDRA. Marketing SMS requires consent evidence, 07:00–21:00 timing and AD-prefixed sender classification.

Limit: Carrier implementation/registration documentation remains operationally provider-specific.

Open source

P12 Marketing SMS FAQs

UAE TDRA. Explicit consent, free unsubscribe, consent record and permitted hours for promotional SMS.

Limit: FAQ wording may summarize rather than replace regulation.

Open source

P13 United Arab Emirates SMS Guidelines

Twilio. Registered alphanumeric sender required; two-way/long/short code unsupported; promotional domestic-only with AD- prefix; URL sensitivity; two-week indicative provisioning.

Limit: Vendor guidance, not law; routes and availability can change without notice.

Open source

P14 How to Register your Alphanumeric Sender ID in UAE

Twilio. New registrations currently on hold; separate transactional/promotional process; international setup USD225 and monthly USD115 fees; documents required.

Limit: Hold/fees may change; confirm with Twilio sales/compliance before plan commitment.

Open source

P15 Requesting publishing access — Loyalty cards

Google for Developers. Demo Mode test-user restriction; Business Profile/payments profile and class required; Google review grants production publishing.

Limit: Review duration/SLA and approval outcome unpublished.

Open source

P16 Retail Payment Services and Card Schemes Regulation — Licensing

Central Bank of the UAE. No person may provide or promote listed Retail Payment Services in UAE without licence unless exempt.

Limit: Whether a specific orchestration model crosses regulated boundary depends on contracts and fund/control flow.

Open source

P17 CBUAE Licensing

Central Bank of the UAE. Current licensed entities list and Retail Payment Service categories, including merchant acquiring and payment aggregation.

Limit: List changes; recheck immediately before PSP contracting.

Open source

P18 Retail Payment Services and Card Schemes Regulation

Central Bank of the UAE. In-force UAE framework covers licensing, AML/CFT, technology/security and user obligations for PSPs.

Limit: Legal perimeter analysis requires UAE counsel.

Open source

P19 Federal Decree-Law No. 45 of 2021 Concerning Protection of Personal Data

UAE Legislation. Federal PDPL principles, controller/processor duties and Articles 22–23 cross-border transfer mechanisms.

Limit: Applicability/exemptions and implementing decisions must be assessed for entity/data context.

Open source

P20 Data Export & Sharing: Transferring Personal Data

DIFC Commissioner of Data Protection. DIFC transfers require adequacy under Article 26 or safeguards/derogations under Article 27.

Limit: Need entity-specific export assessment.

Open source

P21 Guidance on Data Protection Regulations 2021 Part 6: International Transfers

ADGM Office of Data Protection. Foreign access is a transfer; onshore UAE is outside ADGM; no de minimis threshold; adequacy/safeguard/derogation routes.

Limit: Apply current Regulations and adequacy list to actual vendor locations.

Open source

E01 Render regions

Render. Render service and datastore region list; region change requires migration; same-region private networking

Open source

E02 Available regions

Supabase. Supabase primary region model; specific region list; region as data-location control rather than compliance proof

Open source

E03 Change Project Region

Supabase. Supabase region change requires new project and migration

Open source

E04 Connect to Postgres

Supabase. Frontend Data API requires RLS; RLS with no policies denies requests; persistent backend connection choices

Open source

E05 Wallet Developer Guide: Wallet ecosystem and pass updates

Apple. Server sends push notification and device fetches latest pass; updates distribute complete latest pass

Open source

E06 Getting Started with Apple Wallet

Apple. Apple-issued certificate signs passes; certificate expiry/revocation effects; web distribution without related app; pass updates

Open source

E07 Wallet Developer Guide: Pass Design and Creation

Apple. Pass type identifier and certificate relationship; manifest and PKCS#7 detached signature; signed compressed package

Open source

E08 Azure geographies

Microsoft Azure. Azure lists UAE North and UAE Central regions; service availability must be checked per region

Open source

E09 Render pricing

Render. Contractual uptime SLA advertised for Enterprise; plan-dependent HA/PITR and support features

Open source

E10 Service Level Agreement

Supabase. 99.9% uptime commitment for covered Enterprise products; scope, exclusions, and service credits

Open source

E11 Production Checklist

Supabase. PITR option and restore granularity; backup and production configuration considerations

Open source

A01 Apple Developer Program License Agreement, Attachment 5 and section 2.9

Apple Developer. Own-brand pass rule, third-party-mark rights, service-provider exception, no signing third-party passes, licensee accountability

Version: page published 2026-10-26; accessed 2026-10-27

Open source

A02 PassKit Developer Documentation — Certificates API

PassKit. PassKit account can upload/manage Apple pass certificates tied to pass type/team/owner metadata

Version: developer docs v1.1.159 site; accessed 2026-10-27

Limit: Vendor documentation; does not determine Apple-contract compliance or merchant ownership.

Open source

A03 Pricing on the WhatsApp Business Platform

Meta for Developers. Rate-card links, 2026-10-01 service charging and 1,000 free service messages per business phone number/month

Version: page updated 2026-09-30; accessed 2026-10-27

Open source

A04 AED list rates artifact

Meta.

Version: effective 2026-10-01; retrieved 2026-10-27; signed query URL decoded from Meta redirect. Included archive: meta-whatsapp-aed-list-rates-effective-2026-10-01.xlsx. SHA-256: 3a5a0b92fac1f9de229bfadfd2bd5192da46573d5775225e52afc4d3e7b94c47

Open source

A05 USD list rates artifact

Meta.

Version: effective 2026-10-01; retrieved 2026-10-27; signed query URL decoded from Meta redirect. Included archive: meta-whatsapp-usd-list-rates-effective-2026-10-01.xlsx. SHA-256: 3e996b56a86d9e7f70a9cfded896c8f39a19cb0e88683ed06b28f191856c9260

Open source

A06 United Arab Emirates SMS Guidelines

Twilio. Registered alphanumeric sender requirement; one-way limitations; domestic promotional restrictions; URL sensitivity

Version: live guidance accessed 2026-10-27; no visible revision date

Open source

A07 How to Register your Alphanumeric Sender ID in UAE

Twilio Help Center. New-registration hold, documents, domestic/international categories, USD 225 setup and USD 115 monthly international sender fees

Version: live help article accessed 2026-10-27; no visible revision date; fees effective 2025-01-01

Limit: Vendor guidance; hold and fees require written reconfirmation.

Open source

G01 Member States

Cooperation Council for the Arab States of the Gulf — Secretariat General. Official six-member scope: UAE, Bahrain, Saudi Arabia, Oman, Qatar and Kuwait.

Limit: Geography only; no commercial-market evidence.

Open source

G02 Business Demography Statistics 2024

Saudi General Authority for Statistics. 1.022m active enterprises; activity distribution/counts and new/closed enterprise indicators.

Limit: Enterprises are not merchant locations; broad ISIC activities do not isolate target workflows.

Open source

G03 Structural Business Statistics for Wholesale and Retail Trade Activities 2024

Saudi General Authority for Statistics. SAR919bn wholesale/retail operating revenue; SAR410.3bn retail excluding vehicles; SAR203.6bn motor vehicle sale/repair.

Limit: Revenue breadth is not software demand, outlet count or addressable TAM.

Open source

G04 Monsha'at Reports

Small and Medium Enterprises General Authority (Monsha'at). Official sector-report catalogue covers wholesale/retail, accommodation/food and other services, including technology context.

Limit: Listed sector reports vary in age; catalogue does not establish current merchant counts or product demand.

Open source

G05 MCIT Showcases Milestones of the SMEs Go Digital Program

Qatar Ministry of Communications and Information Technology. Programme sector scope; 173 SMEs onboarded, 94 matchmaking sessions, 59 providers and 30 adopted digital solutions as reported for 2025.

Limit: Programme outputs, not national adoption or willingness-to-pay.

Open source

G06 The National E-commerce Strategy

Oman Ministry of Transport, Communications and Information Technology. Official strategy pillars include SME enablement, payments, logistics, regulation, skills and digital commerce.

Limit: Policy intent; no target-sector demand or merchant purchasing evidence.

Open source

G07 National Digital Economy Strategy 2025

Bahrain Information & eGovernment Authority. Official strategy includes digital solutions for SMEs among transformative initiatives.

Limit: Policy direction; no merchant count, budget or adoption evidence.

Open source

G08 Salla Plans and Unified Commerce for Retail

Salla. First-party Saudi commerce plans, order/payment features and pricing; retail proposition links online and physical retail/POS.

Limit: Vendor claims; entitlements can change; no proof of merchant outcomes or neutral handoff interoperability.

Open source

G09 Zid Pricing

Zid. First-party Saudi pricing and advertised e-commerce, POS, WhatsApp, inventory/branch and API features by tier.

Limit: Promotions/entitlements can change; integration and partner rights require confirmation.

Open source

G10 Fresha Pricing and Product Scope

Fresha. Bookings, reminders, deposits/prepayment, client management, team calendars and supported beauty/wellness categories.

Limit: First-party global page; does not verify country-specific GCC contracting, availability or price.

Open source

G11 Queueing Solution & Service Intelligence Pricing

Qminder. Queue/check-in, messaging, appointments, analytics, API/webhooks, security scope and public USD annual prices.

Limit: Global first-party pricing, not a GCC local-currency offer; hardware, implementation and messaging may add cost.

Open source

G12 Queberry Queue Management

Queberry. First-party UAE/regional queueing, WhatsApp, signage and analytics positioning.

Limit: No official public price or independently verified outcomes.

Open source

G13 Foodics UAE Pricing and Digital Receipt Integration

Foodics. OAuth and customer-order webhook receipt path; combined with documented UAE commercial presence already reviewed in market workstream.

Limit: Does not prove custom print-template QR, all-order coverage or kitchen-ready events; pricing is on a separate first-party page.

Open source

G14 Annual Dubai Gastronomy Industry Report 2024

Dubai Department of Economy and Tourism. Official Dubai restaurant-licence, dining-frequency, tourism and food-service activity signals used in existing market analysis.

Limit: Restaurant licences are not active café/QSR locations; consumer survey does not show merchant willingness-to-pay.

Open source

G15 86.7% Completion Rate for the National E-Commerce Plan (2022–2027)

Oman Ministry of Commerce, Industry and Investment Promotion. Official report of 19,260 licensed e-commerce activities during stated 2024 window and 412 Maroof Oman verified stores.

Limit: Activity/licence definition and stock-versus-flow interpretation are unclear; figures are not target merchant locations.

Open source

C01 Personal Data Protection Law

Saudi Data & AI Authority (SDAIA). Saudi PDPL controller duties and Article 29 conditions for transfers outside Kingdom.

Limit: Entity-specific scope, legal basis and exceptions require Saudi counsel.

Open source

C02 Regulation on Personal Data Transfer outside the Kingdom (official knowledge-center text)

Saudi Data & AI Authority / National Data Management Office. Articles 2–7 cover transfer purposes, adequacy assessment, safeguards/exemptions, onward transfers and specified risk-assessment cases.

Limit: Arabic official portal text verified; current adequacy list and chosen safeguard must be checked for each vendor destination.

Open source

C03 E-Invoicing Implementation Resolution — controls, requirements and technical specifications

Saudi Zakat, Tax and Customs Authority (ZATCA). Simplified electronic invoices require ZATCA-specified QR; compliant solution generates invoice fields and QR.

Limit: Arabic text controls on discrepancy; taxpayer wave and POS implementation need merchant tax adviser.

Open source

C04 Regulations of the Provision of the Short Messaging Service SMS

Saudi Communications, Space & Technology Commission (CST). 2024 SMS service regulation and permit framework; service covers provider/customer-directed promotional and informational messaging.

Limit: Does not itself prove merchant sender-name approval or platform exemption; confirm with permitted provider/CST.

Open source

C05 Executive Regulations of the Personal Data Protection Law — Ministerial Decision 34/2024

Oman Ministry of Transport, Communications and Information Technology (MTCIT). Controller/processor duties, Article 5 permits, child data, breach, DPO, marketing and overseas-transfer controls under Royal Decree 6/2022.

Limit: Permit and transfer requirements must be applied to actual data classes and processors.

Open source

C06 Order concerning countries and territories with adequate protection and transfer authorization

Bahrain Personal Data Protection Authority. Transfers outside listed adequate territories can require case-by-case Authority authorization; includes group-transfer process.

Limit: Check current adequate-country record and any later orders at launch.

Open source

C07 Guidance Hub for Regulated Entities — PDPPL

Qatar National Cyber Security Agency (NCSA). Law 13/2016 PDPPL applies to electronically processed personal data; official controller, processor, privacy-by-design, DPIA, breach, rights and direct-marketing guidance.

Limit: Organization-specific measures and cross-border position require local assessment.

Open source

C08 Decision 26/2024 issuing Data Privacy Protection Regulation

Kuwait Communication and Information Technology Regulatory Authority (CITRA). Decision 26/2024 issues current regulation and repeals Decision 42/2021 and amendments.

Limit: CITRA statutory/regulatory scope and interaction with other Kuwaiti laws need counsel; do not describe as blanket omnibus PDPL.

Open source

C09 Global availability

Stripe. UAE is listed for Stripe business account support; other five GCC states are not listed.

Limit: Availability and product scope can change; recheck at contracting.

Open source

C10 Requirements to open a Stripe account in another country

Stripe Support. Supported foreign-country account needs same-country legal entity, tax ID, physical location and physical bank account, plus other verification items.

Limit: Underwriting and product eligibility remain account-specific.

Open source

C11 Getting Started with Apple Wallet

Apple Developer. PassKit pass distribution/signing model and requirement to test barcode scanner hardware.

Limit: Page does not promise every pass feature on every GCC device/account.

Open source

C12 Supported countries for Google Wallet and tap to pay

Google Wallet Help. Country selector includes all six GCC states; Wallet app can store passes/cards while specific items and payment methods vary by country, device and partner.

Limit: Country listing is not issuer, bank, pass-type or production-publishing approval.

Open source

C13 Complying with regional regulations

Apple Developer. Saudi domestic debit requires mada support; UAE domestic debit/prepaid requires Jaywan support; PSP support must be confirmed.

Limit: Payment acceptance still depends on merchant PSP, bank, device and card.

Open source

C14 Current Currency & Funds — List One

SIX Group — ISO 4217 Maintenance Agency. Official current ISO 4217 code and minor-unit list: AED/SAR/QAR use 2; BHD/KWD/OMR use 3.

Limit: Recheck current XML/XLS during implementation upgrades.

Open source

C15 WhatsApp Business Platform pricing and country calling-code market mapping

Meta for Developers. Effective-date rate-card model and market mapping; local archived AED workbook supplies exact 2026-10-01 GCC rates.

Limit: Rate cards are mutable; Bahrain is mapped to workbook region rather than a named row, so revalidate mapping before billing.

Open source

C16 Generic Private Pass overview

Google for Developers. Sensitive-data passes need explicit Google permission, verification and extra privacy review.

Limit: Approval does not establish compliance with local health-data law.

Open source

R01 Web Push for Web Apps on iOS and iPadOS

WebKit. Home Screen web app push and permission conditions; QR page alone does not imply background alert entitlement.

Open source

R02 The Goal Gradient Hypothesis Resurrected

Journal of Marketing Research and author university repository. 2006 cafe reward research on goal proximity and purchase acceleration; not a UAE profit estimate.

Open source

R03 The Endowed Progress Effect

Journal of Consumer Research. Artificial initial progress with unchanged remaining requirements can affect persistence; moderators and context matter.

Open source

R04 The Design and Targeting of Compliance Promotions

Marketing Science. Participation selection, randomized promotion evaluation and incremental margin after reward costs.

Open source

R05 API Broken Object Level Authorization

OWASP. Object authorization across APIs; opaque identifiers do not replace access controls.

Open source

R06 Web Content Accessibility Guidelines 2.2

W3C. Accessibility acceptance reference; testing target is proposed, not certification.

Open source

R07 UAE Pricing and Fees

Stripe. Domestic payment processing 2.9 percent plus AED1 and recurring Billing 0.7 percent; one-off invoices excluded from Billing volume.

Open source

R08 Pricing and Fees

Supabase. Pro starting price, included project/compute credit and separately priced PITR; not a capacity guarantee.

Open source

R09 Production Rails Hosting Architecture and Scaling Guide

Render. September 2026 published example with US$7 web and US$7 worker processes; reference price only.

Open source

R10 Use Loopy Loyalty as a Managed Service

Loopy Loyalty. Managed-service setup and recurring charging permitted; not an automatic full white-label entitlement.

Open source

R11 Introduction to PassKit Partnerships

PassKit. Integration/referral paths distinguished from resale; vendor says no reseller program.

Open source

R12 Trigger Push Notifications for Loyalty Cards

Google for Developers. Loyalty-specific notification flags, limits, permissions and nearby behavior.

Open source

R13 Introduction to Manufacturing Systems Queues

MIT OpenCourseWare. Steady-state Little law and capacity constraints; visibility alone does not add preparation capacity.

Open source

R14 What Types of Businesses Can Open a UAE Account

Stripe Support. UAE business licence/permit and eligible business structures; no assumed account approval.

Open source

R15 What Are Your Pricing Plans

Loopy Loyalty Help Center. Staff limits, higher-tier data export and unique enrollment; costs and fair-use constraints.

Open source